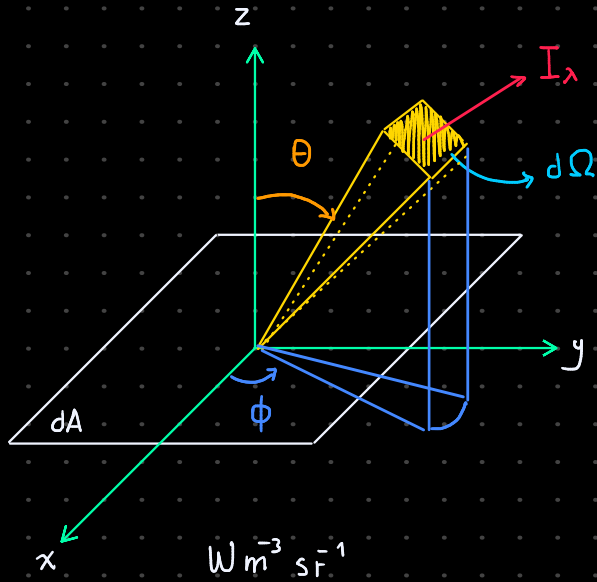


rayo de luz $\lambda, \lambda + d\lambda$



$$dA \cos \theta, \quad E_\lambda \equiv \frac{\partial E}{\partial \lambda}$$

$E_\lambda d\lambda$ cantidad de energía que un rayo lleva dentro del cono en un tiempo dt .

Intensidad específica

$$I_\lambda \equiv \frac{\partial I}{\partial \lambda} \equiv \frac{E_\lambda d\lambda}{d\lambda dt dA \cos \theta d\Omega}$$

$\lambda, \lambda + d\lambda$

$$\rightarrow E_\lambda d\lambda = \overset{\text{W m}^{-2} \text{ sr}^{-1}}{I_\lambda} d\lambda dt dA \cos \theta d\Omega = I_\lambda d\lambda dt dA \cos \theta \sin \theta d\theta d\phi$$

La función de Planck es un caso especial de I_ν

$$\bullet B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}$$

$$\bullet B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1}$$

• Intensidad media

$$\langle I_\lambda \rangle = \frac{1}{4\pi} \int I_\lambda d\Omega = \frac{1}{4\pi} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I_\lambda \sin \theta d\theta d\phi$$

isotropico $\langle I_\lambda \rangle = I_\lambda$

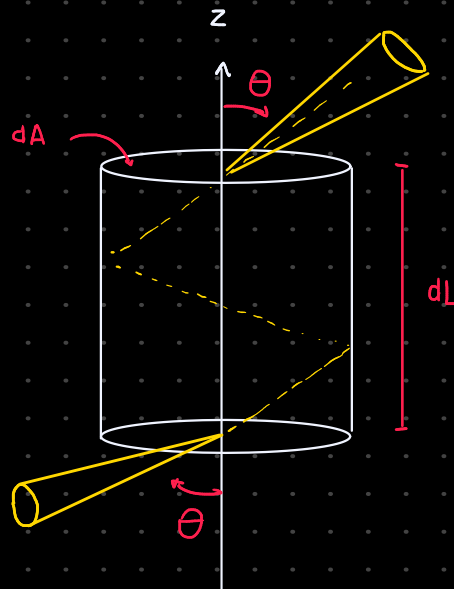
blackbody $\langle I_\lambda \rangle = B_\lambda$

• densidad de energía específica

$$dt = \frac{dL}{c \cos \theta}$$

$$E_\lambda d\lambda = I_\lambda d\lambda dt dA \cos \theta d\Omega$$

$$= I_\lambda d\lambda dA d\Omega \frac{dL}{c}$$



$$dA dL = \text{volumen}$$

$$\begin{aligned} \frac{E_\lambda d\lambda}{dA dL} &= u_\lambda d\lambda = \frac{1}{c} \int I_\nu d\lambda d\Omega \\ &= \frac{1}{c} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} I_\lambda d\lambda \sin\theta d\theta d\phi \\ &= \frac{4\pi}{c} \langle I_\lambda \rangle d\lambda \end{aligned}$$

• isotrópico $u_\lambda d\lambda = \frac{4\pi}{c} I_\nu d\lambda$

• blackbody $u_\lambda d\lambda = \frac{4\pi}{c} B_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{e^{hc/\lambda k_B T} - 1}$

$$u = \int_0^\infty u_\lambda d\lambda = \int_0^\infty u_\nu d\nu$$

• blackbody $u = \frac{4\pi}{c} \int_0^\infty B_\lambda(T) d\lambda = \frac{4\sigma T^4}{c} = a T^4$

$$a \equiv \frac{4\sigma}{c} = 7.566 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4} \quad \text{constante de radiación}$$

• Presión de radiación

$$\boxed{F = \frac{dp}{dt}} = \frac{d}{dt} (mv) = \frac{dm}{dt} v + m \cdot \frac{dv}{dt}$$

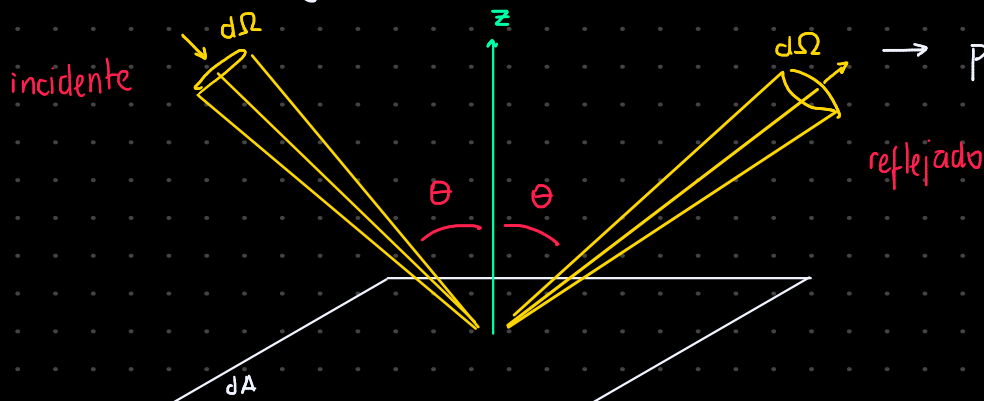
$$P = \frac{F}{A} \Rightarrow \frac{\frac{dp}{dt}}{dA}$$

• Energía relativista

$$E^2 = p^2 c^2 + \cancel{m^2} c^4$$

$$E^2 = p^2 c^2$$

$$\rightarrow P = \frac{E}{c}$$



$$\begin{aligned}
 dp_{\lambda} d\lambda &= \left[(p_{\lambda})_{\text{final},z} - (p_{\lambda})_{\text{initial},z} \right] d\lambda \\
 &= \left[\frac{E_{\lambda}}{c} \cos\theta - \left(-\frac{E_{\lambda}}{c} \cos\theta \right) \right] d\lambda \\
 &= 2 \frac{E_{\lambda}}{c} \cos\theta d\lambda
 \end{aligned}$$

$$dp_{\lambda} d\lambda = \frac{2}{c} I_{\lambda} d\lambda dt dA \cos^2\theta d\Omega$$

$$\frac{dp_{\lambda} d\lambda}{dt dA} = \frac{2}{c} I_{\lambda} d\lambda \cos^2\theta d\Omega$$

$$P_{\text{rad},\lambda} d\lambda = \frac{2}{c} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I_{\lambda} d\lambda \cos^2\theta \sin\theta d\theta d\phi$$

$$\hookrightarrow \text{total} \quad P_{\text{rad},\lambda} = \frac{4\pi}{3c} I_{\lambda} d\lambda$$

$$P_{\text{rad}} = \int_0^{+\infty} P_{\text{rad},\lambda} d\lambda$$

$$\bullet \text{ black body} \quad P_{\text{rad}} = \frac{4\pi}{3c} \int_0^{+\infty} B_{\lambda} d\lambda = \frac{4\pi}{3c} \sigma T^4 = \frac{1}{3} a T^4 = \frac{1}{3} u$$

Equilibrio hidrostático

$$\frac{dP}{dr} = -\rho \frac{GM(r)}{r^2}$$

EoS $\rightarrow$ gas ideal

$$PV = Nk_B T$$

$$P = n k_B T$$

$$\rho = \rho_H + \rho_{He} + \rho_{metals} \quad \left| \quad X + Y + Z = 1 \right.$$

$$\hookrightarrow P = \frac{\rho k_B T}{\mu m_u}$$

mean molecular weight

• composición

• ionización

mass fraction H $X \equiv \frac{\rho_H}{\rho}$

He $Y \equiv \frac{\rho_{He}}{\rho}$

metals $Z = \frac{\rho_{metals}}{\rho} = 1 - X - Y$

• Fotosfera del Sol: $X_0 = 0.734$, $Y_0 = 0.250$, $Z_0 = 0.016$

• Hidrógeno

$\rightarrow$ neutro $\mu = 1$, $n = \frac{\rho}{m_p}$



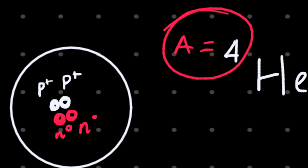
$$n = \frac{\rho}{\mu m_p}$$

$\rightarrow$ ionizado $\mu = \frac{1}{2}$, $n = \frac{2\rho}{m_p}$



• Helio

$\rightarrow$ neutro $\mu = 4$, $n = \frac{\rho}{4m_p}$



$\rightarrow$ fully ionised $\mu = \frac{4}{3}$, $n = \frac{3\rho}{4m_p}$



• Metales

$\rightarrow Z = e^-$

A mass number, $A = Z + n^0$, isótopos estables $\sim A/2$ electrones

$\rightarrow$ neutro $\mu = A$, $n = \frac{\rho}{Am_p}$

$\rightarrow$ fully ionised $\mu = 2$, $n \approx \frac{\rho}{2m_p}$

ionizado

$$A \geq Z$$

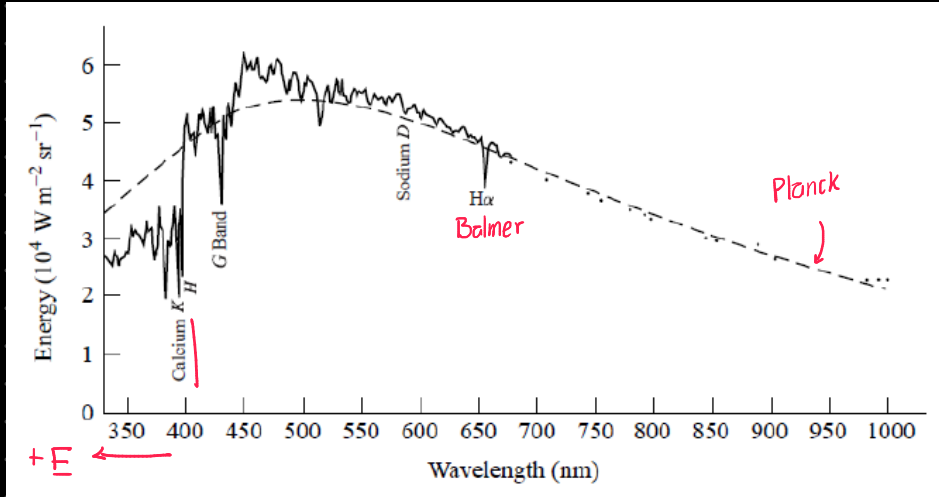
$$n \approx X \left(\frac{2p}{m_p} \right) + Y \left(\frac{3p}{4m_p} \right) + Z \left(\frac{p}{2m_p} \right)$$

$$n = \left(2X + \frac{3}{4}Y + \frac{Z}{2} \right) \frac{p}{m_p} \longrightarrow \mu_{\text{ionizado}} = \frac{p}{n m_p} = \left(2X + \frac{3}{4}Y + \frac{Z}{2} \right)^{-1}$$

neutro

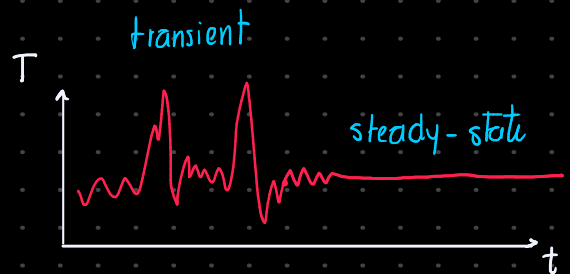
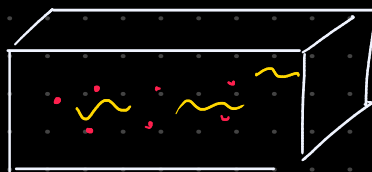
$$n = \left(X + \frac{Y}{4} + \frac{Z}{A} \right) \frac{p}{m_p} \longrightarrow \mu_{\text{neutro}} = \frac{p}{n m_p} = \left(X + \frac{Y}{4} + \frac{Z}{A} \right)^{-1}$$

• Opacidad estelar



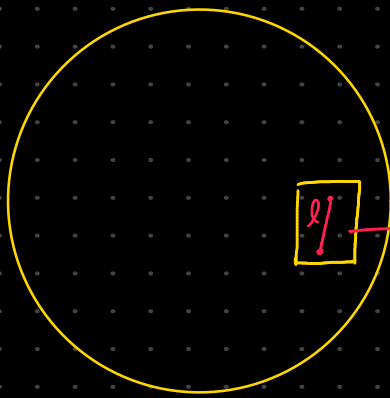
line blanketing

- Temperatura efectiva (Stefan - Boltzmann) ✓
- Temperatura de excitación (ec. de Boltzmann)
- Temperatura de ionización (ec. de Saha)
- Temperatura cinética (dist. Maxwell - Boltzmann)
- Temperatura de color (fitting del espectro a la función de Planck)



equilibrio termodinámico

• ionización $\rightleftharpoons$ recombinación



LTE

mean free path

$$\ell = \frac{1}{n\sigma}$$

- Maxwell - Boltzmann
- Ec. Boltzmann
- Saha